

Roll No. 129194072

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

B. Tech. (Semester - IV) Theory Examination
Subject: Numerical and Statistical Methods (For EE)

Paper Code: MAIC-203

Time: 03 Hours

Max. Marks: 50

Note:

- I. Answer ALL questions. Marks allotted for each question are shown on the right hand margin.
- II. The candidates, before starting to write the solutions, should please check the question paper for any discrepancy, and also ensure that they have been delivered the question paper of right course number and the right course title.
- III. Unless stated otherwise, the symbols have their usual meanings in context with the subject.

1	(a) Find by the Newton-Raphson method, the real root of the equation $3x = \cos x + 1$, correct to four decimal places. [5] (b) Solve the following equations by Gauss-Seidal method: [5] $\begin{aligned} 5x + y + 2z &= 19 \\ x + 4y - 2z &= -2 \\ 2x + 3y + 8z &= 39 \end{aligned}$																													
2	(a) The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface: [5] <table> <tr> <td>x (height):</td> <td>100</td> <td>150</td> <td>200</td> <td>250</td> <td>300</td> <td>350</td> <td>400</td> </tr> <tr> <td>y (distance):</td> <td>10.63</td> <td>13.03</td> <td>15.04</td> <td>16.81</td> <td>18.42</td> <td>19.90</td> <td>21.27</td> </tr> </table> Find the value of y at $x = 218$ ft using Newton's forward interpolation. (b) A Solid of revolution is formed by rotating about x-axis, the area between the x-axis, the lines $x = 0$ and $x = 1$ and a curve through the points with the following co-ordinates [5] <table> <tr> <td>x:</td> <td>0.00</td> <td>0.25</td> <td>0.50</td> <td>0.75</td> <td>1.00</td> </tr> <tr> <td>y:</td> <td>1.0000</td> <td>0.9896</td> <td>0.9589</td> <td>0.9089</td> <td>0.8415</td> </tr> </table> Estimate the volume of the solid formed using Simpson's rule.	x (height):	100	150	200	250	300	350	400	y (distance):	10.63	13.03	15.04	16.81	18.42	19.90	21.27	x:	0.00	0.25	0.50	0.75	1.00	y:	1.0000	0.9896	0.9589	0.9089	0.8415	
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3	(a) Solve the following by Euler's modified method: [5] $\frac{dy}{dx} = \log(x + y), \quad y(1) = 2 \text{ at } x = 1.2 \text{ and } 1.4 \text{ with } h = 0.2.$ (b) Using the finite difference method, find $y(0.25)$, $y(0.5)$ and $y(0.75)$ [5] satisfying the differential equation $\frac{d^2y}{dx^2} + y = x$, subject to the boundary conditions $y(0) = 0, y(1) = 2$.																													

4	(a) A curve passes through the points (0, 18), (1, 10), (3, -18) and (6, 90). Find the slope of the curve at $x = 2$ using Lagrange's formula. [5] (b) Using Runge-Kutta method of order 4, find y for $x = 0.1$ and 0.2 given that $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$ by taking $h = 0.1$. [5]
5	(a) A has one share in a lottery in which there is 1 prize and 2 blanks; B has three shares in a lottery in which there are 3 prizes and 6 blanks; compare the probability of A's success to that of B's success. [5] (b) A box A contains 2 white and 4 black balls. Another box B contains 5 white and 7 black balls. A ball is transferred from the box A to the box B. Then a ball is drawn from the box B. Find the probability that it is white. [5]

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